

ABHINAV BIJU

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[Linkedin](#) | [HackerRank](#) | [Portfolio](#) | [GitHub](#) | [LeetCode](#)

EDUCATION

Vellore Institute of Technology (VIT)
Computer Science (Core) B.Tech

Chennai, India
July 2026 - Present

Indian Institute of Technology (IIT)
Data Science and Applications B.S (online)

Chennai, India
May 2026 - Present

SKILLS

Programming Languages: Java, Python, C/C++
Libraries/Frameworks: JavaScript
Tools / Platforms: VS Code, Claude, Render, Docker, VirtualBox, Flask, Git
Databases: Supabase (PostgreSQL)

PROJECTS / OPEN-SOURCE

SkynetGrid | [Link](#) *Java, TCP/IP, UDP, Java Swing, Multithreading, Linux/systemd*
A **distributed Java remote-administration system** built around a server hub, lightweight LAN clients, and a Swing-based administration console communicating through a custom TCP/UDP protocol. Implements **automatic server discovery and election**, live screen streaming with change detection, remote mouse and keyboard control, file management, terminal command execution, and asynchronous request handling. The system also includes self-healing `systemd` deployment and was iterated through **34+ development versions** before being deployed with school approval.

Greed-Island | [Link](#) *Java, Multithreading, Genetic Algorithms, Probability & Statistics*
A **multithreaded evolutionary simulation** in Java featuring 20 autonomous agents, each operating as an independent thread and making probabilistic decisions based on inherited traits, environmental conditions, and internal state. Agents compete for limited resources, survive, reproduce, and pass mutated traits to subsequent generations, allowing complex survival strategies to emerge without explicitly programming their behavior. Implements **genetic inheritance, adaptive mutation, weighted probability, concurrent agent execution, resource competition, and event-driven simulation logic**.

Grimoire-Legacy | [Link](#) *Java, OOP, Data Structures & Algorithms, Design Patterns*
A **Java turn-based spell-combat engine** evolved from a pen-and-paper game into a complex simulation supporting **150+ spells and 30+ AI opponents**. Implements simultaneous turn resolution, cooldowns, stacking effects, damage-over-time mechanics, and a linked-list-based turn queue for managing combat state. Uses **polymorphism, shared spell definitions, Prototype-based state cloning, priority queues, and custom effect-resolution logic** to manage interactions between spells, entities, status effects, and AI-controlled opponents.

FileDashboard | [Link](#) *Java, JDK HTTP Server, HTML, CSS, JavaScript, Batch*
A **Java-based browser-accessible file management system** built using the JDK's built-in HTTP server without external frameworks or build tools. Provides a web interface for **directory navigation, recursive file search, tabbed browsing, file previews, text/Markdown/PDF viewing, file operations, ZIP handling, and recycle-bin management**. Also implements activity-based file ranking, configurable settings, and optional Windows autostart, combining a custom Java backend with a browser-based frontend to create a lightweight local file explorer.

Navigo | [Link](#) *Python, PyQt5, QtWebEngine, Chromium*
A **Python desktop web browser** built with PyQt5 and QtWebEngine, providing a custom browser interface and application layer around Qt's Chromium-based rendering engine. Implements **tabbed browsing, multiple windows, incognito mode, bookmarks, browsing history, search suggestions, downloads, context menus, keyboard shortcuts, and event-driven UI interactions**. The project focuses on building the browser's surrounding application architecture and user experience while leveraging QtWebEngine for standards-compliant web rendering.

Mining-Simulator | [Link](#)

Java, File I/O, GCD/LCM Algorithms, Weighted Randomization

A **Java console-based mining and collection simulator** featuring persistent user accounts, inventories, events, and **35 minerals** distributed across three collections and four rarity variants. Designed a dynamic weighted-random drop system using **GCD/LCM-based probability pools**, converting arbitrary rarity ratios into integer weights without manually recalculating individual probabilities when the system changes. Also implements custom file-based persistence for accounts and inventory state, alongside rarity-based events and collection progression.

VisualCanvas | [Link](#) *Java, Linux, Data Structures, Physics Simulation, Computational Geometry, File IO*

A **Linux-based interactive visualization and simulation environment** integrating computer science, chemistry, and physics into a unified editor. The CS canvas provides interactive representations of **linked lists, stacks, queues, heaps, trees, BSTs, arrays, and graphs**, with support for custom data values. The chemistry module includes a **molecule loader with a collection of prebuilt molecular structures**, allowing users to load existing molecules and use built-in **editing tools to modify and work with molecular models**. Alongside this, the application features a **high-resolution optimized physics engine** supporting constraints, pulleys, springs, ropes, rods, and gravitational interactions while handling large numbers of objects. Projects can be edited and persisted using the custom `.vcanvas` file format, with support for saving, loading, and workspace screenshot export.

CERTIFICATIONS

- Microsoft Office Specialist: PowerPoint (Office 2016) - **Microsoft**
- Java (Basic) - **HackerRank**

HONORS & AWARDS

- Won National 1st Runner up at ICT JUNIOR INDIA CHAMPIONSHIP (Powerpoint 2016)